SHIFTING TONES IN HISTORICAL POSTCARDS

A distance viewing analysis of Erlangen (1897-1921)

Friedrich-Alexander-Universität Erlangen-Nürnberg

Faculty of Natural Science

Department of Data Science

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Dr. Sabine Lang

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Author : Srinithi Badrinath

23698700

srinithi.badrinath@fau.de

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1. Introduction

This presentation investigates how the colour and tonal properties of historical postcards from Erlangen evolved between 1897 and 1921. Using the Distant Viewing Explorer tool, the study analyses colour feature such as brightness, saturation, edge score, and hue distribution. The goal was to identify large-scale aesthetic patterns across time.

2. Dataset Overview

Postcards are a special kind of everyday visual communication. Unlike paintings or photographs, postcards were designed for mass consumption and circulated widely. Postcards capture how a city wanted to present itself to visitors and residents, offering insight into aesthetic and cultural shifts.

The dataset consists of 50 digitized historical postcards of Erlangen spanning over the period from 1897 to 1921 covering the transition from late 19th-century monochrome or lightly tinted postcards to early 20th-century colour-printed editions. The images are collected from Wikimedia Commons which is a public digital library. The metadata was created manually with information such as filename, description, dimensions, category, year, source and the image. The postcards depict a wide range of subjects, including architectural landmarks, cityscapes, railway stations, markets, festivals and public squares.

Filename	Description	Dimension	Category	Year	Source	Image
Erlangen_-_Bahnhof_Postkarte_002	Bahnhof Erlangen	1,000 × 646	Buildings / Streets / Architecture	1899	Wikimedia Commons	
Bergkirchweih_Erlangen_1900_001	Postcard showing the Erlangen Bergkirchweih (traditional beer festival), created around 1900. This likely depicts a festival scene from the turn of the 20th century	5,856 × 3,707	Festivals / Events	1900	Wikimedia Commons	
Ldmk_Schleuse_89	Ludwig-Donau-Main-Kanal, Schleuse 89, Erlangen	1,198 × 750	Nature / Parks / Landscapes	1900	Wikimedia Commons	

Fig 1. Sample shot of the metadata xlsx

3. Analytical Focus

"How do colour and tonal patterns in Erlangen postcards change over time?"

This question focuses on measurable visual characteristics rather than interpretive symbolism. The aim is not to explain why specific images were produced but to observe patterns in appearance that may reflect broader aesthetic developments.

4. Methodology

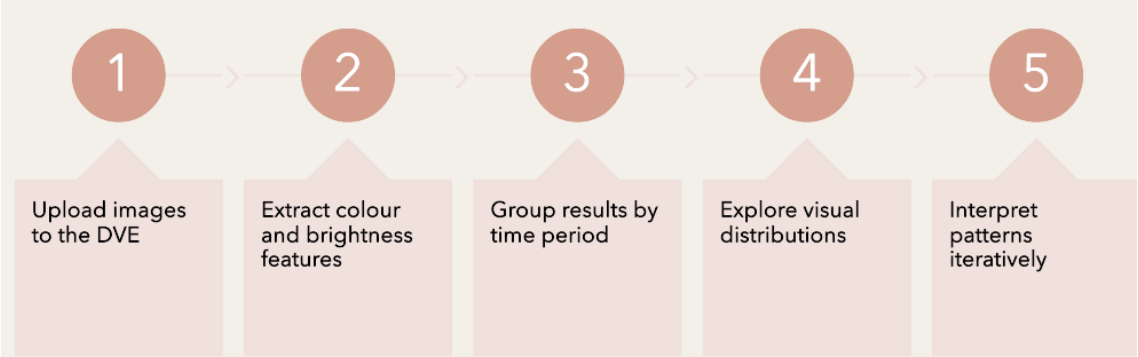

Fig 2. Flowchart depicting the workflow

The analysis was conducted using Distant Viewing Explorer, a free, open-source, web-based software. The following steps were performed:

1. The dataset was imported into the tool.
2. The software generated numerical outputs for:
    a. Brightness (lightness of the image) and Saturation (intensity of colour)
    b. Edge score (an indicator of visual complexity or detail)
    c. Hue composition (percentage of pixels in each hue category: red, orange, yellow, green, blue, purple, black, white, grey)
3. Images were grouped into subsets to examine how visual properties shift across decades.
4. The extracted values were exported, reviewed, and interpreted through charts and descriptive comparisons.

This exploratory methodology emphasizes patterns rather than causal interpretations and aims to reveal macro-level tendencies within the dataset.

5. Analysis

5.1 Brightness, Saturation, and Edge Score

Analysis of the brightness reveals a constant increase over time. In the earlier period, the postcards appeared to be dark and tonally muted, which reflects the monochrome or tinted printing techniques. From around 1905 onward, postcards became noticeably brighter, indicating development in colour printing and higher production quality.

Saturation values also increase slightly over time but remain highly variable depending on subject matter. Festival scenes and landscapes have high saturation, while architectural postcards remain more muted.

Edge score, which represents visual complexity, remains rather unchanged across the dataset. However, there is a slight increase in edge score in later postcards, possibly due to sharper printing methods or more detailed illustrations.

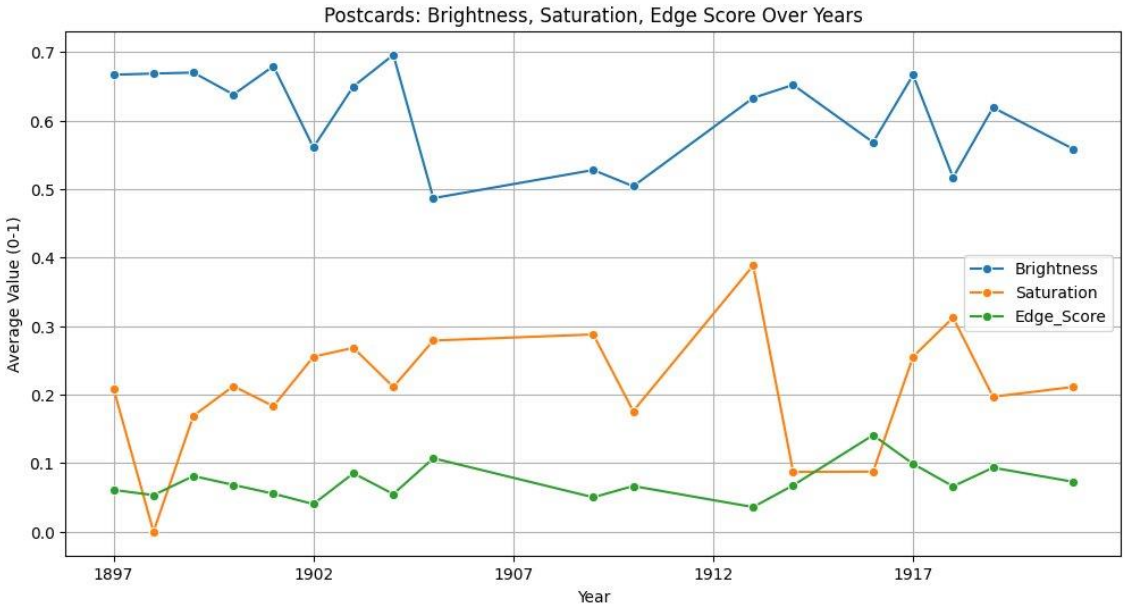

Fig 3. Line Chart comparing Brightness, Saturation and Edge score

5.2 Hues

Evidence of changes over time is more apparent from a Hue analysis. Pre-1905 postcards are dominated by shades of white, black, and grey that suggest either monochromes or hand-coloured with whitewash.

In the mid-periods, the postcards show a significant rise in yellow and orange hues, which appear in skies, buildings, and foliage tinting. These changes occurred with the advent of chromolithographic colour printing.

The postcards from the later period (1915–1921) exhibit more vivid colours, with the intensity of green, blue, and red colours rising, which shows the use of multi-colour printing techniques. Overall, the postcards show a transition from neutral tones to warm, multi-coloured compositions.

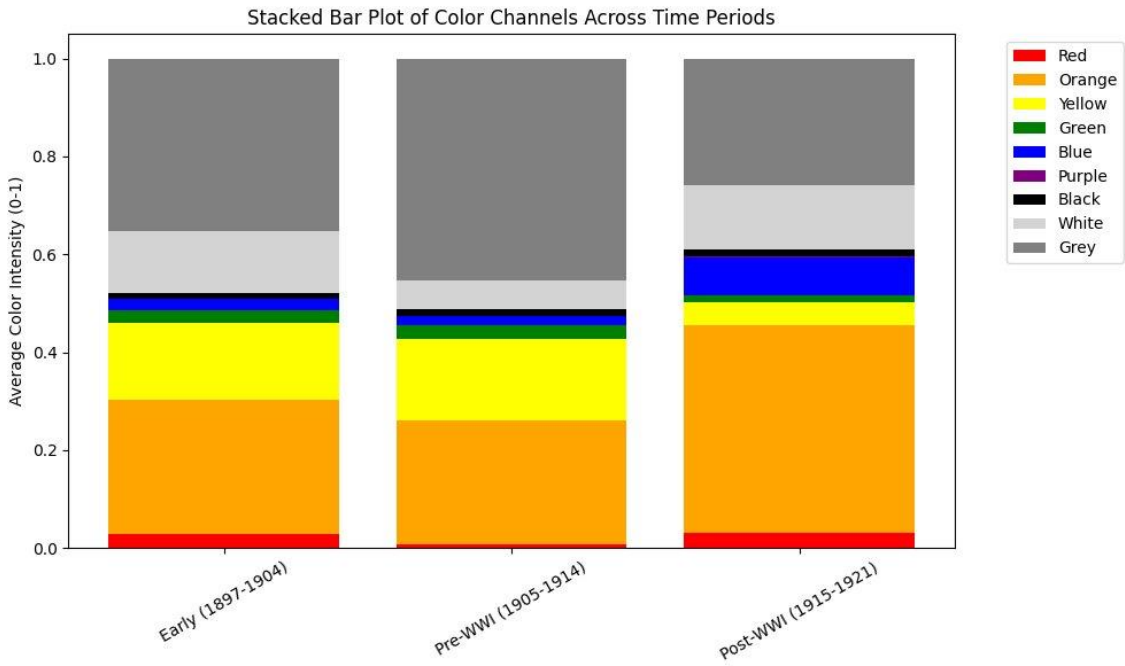

Fig 4. Stacked bar chart indicating Colour channels

5.3 Similarity Through Time

Visual similarity analysis using Embeddings in DVE shows that early postcards cluster closely together, forming tight groups with similar brightness and neutral colour tones. Mid-period postcards show greater diversity, while late-period postcards form several clusters based on colour intensity and subject type.

The similarity timeline suggests that early postcards were more stylistically homogeneous, while later postcards exhibit more visual variety. This may reflect both the increasing availability of colour printing and evolving commercial or artistic preferences in postcard production.

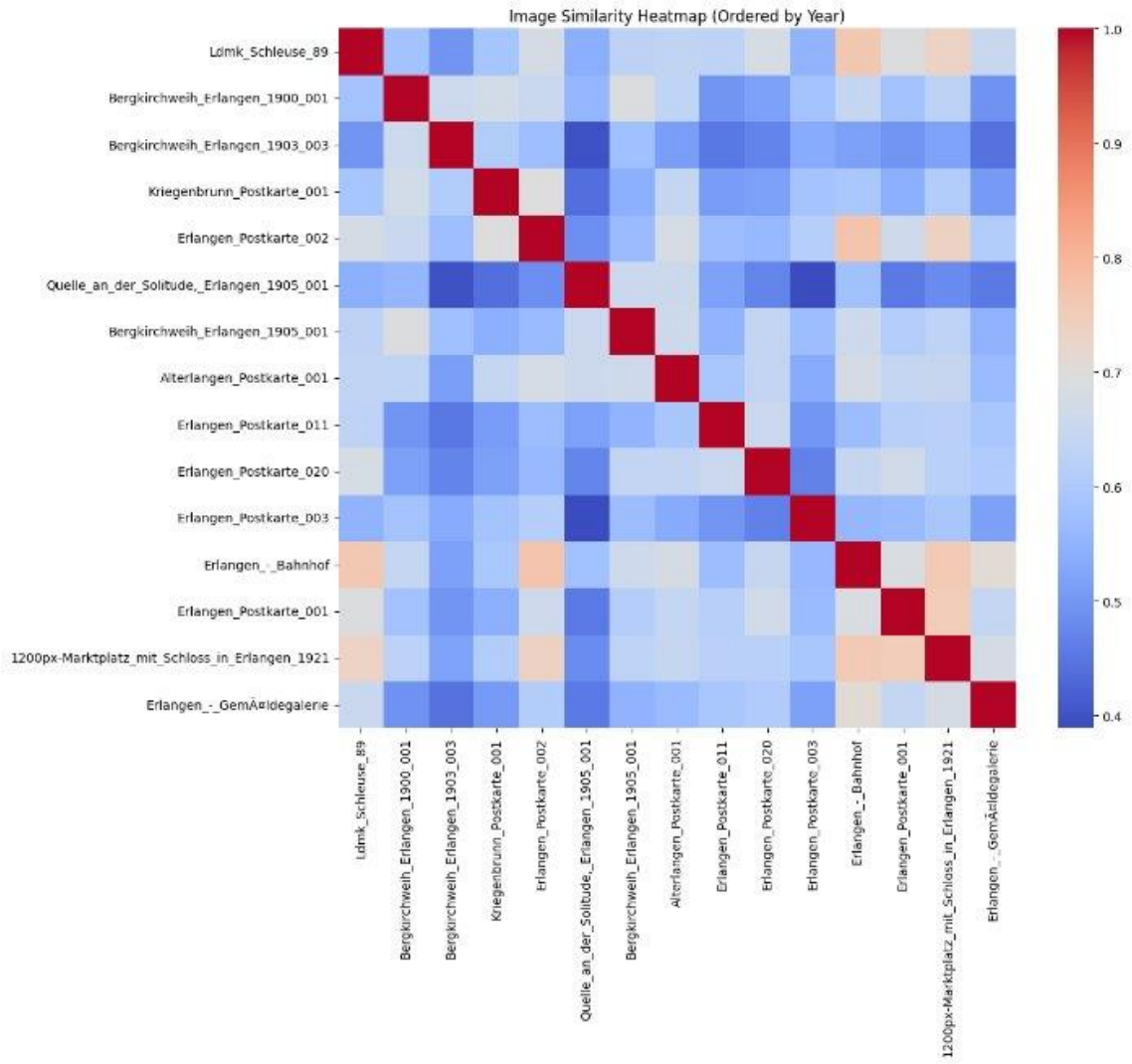

Fig 5. Heatmap showing Image similarity

6. Limitations

Several limitations should be acknowledged:

- Small sample size limits generalizability and statistical confidence.
- Digitization differences (scanning quality, lighting conditions) may affect brightness and colour values.
- Metadata inconsistencies or inaccuracies may lead to incorrect time grouping.
- The analysis considers visual features only, without examining cultural, economic, or artistic contexts.
- No cross-city comparison is included, so changes cannot be attributed solely to Erlangen or postcard industry trends.
- Early postcards may have faded over time, reducing original colour fidelity.

7. Conclusion

This presentation illustrates how computational visual analysis can reveal meaningful patterns in historical image collections. The study finds consistent temporal shifts in the visual characteristics of Erlangen postcards from 1897 to 1921.

Brightness and saturation tend to increase with time, and analysis of hue shows a strong shift from a balanced neutrality in early postcards to more colourful and warmer-hued postcards in the latter years. Similarity clustering also indicated a homogeneous style in earlier postcards and a divergence in styles in later years.

These results demonstrate the importance of distant viewing as a way to comprehend the aesthetic and technological development that has occurred in visual culture over time. Further studies might benefit by using larger data sets, semantic content analysis, or inter-city comparative analytical methods in examining the production of postcards in this historical era.

Declaration of Originality

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